MANUSCRIPT DRAFT · HYDROLOGY AND EARTH SYSTEM SCIENCES (TARGET)

Diagnosing snow-related structural limitations of the Xinanjiang model: a parsimonious snow extension and a two-basin engineering pilot

Working title (avoid "global / first" until stratified large-sample results exist). Status: pilot evidence complete large-sample pending

Authors: *to be completed* · Affiliations: *to be completed* · Correspondence: *to be completed*

Software context: hydromodel v0.3.2; model names XAJ-MZ and XAJ-Snow; Caravan CAMELS subsets.

Contents

Abstract

Conceptual rainfall–runoff models remain central to large-sample hydrology, yet structural adequacy can collapse in regimes that the model never represents. The Xin′ anjiang model (XAJ) in its Muskingum–Zhao routing form (XAJ-MZ) treats precipitation as liquid input and therefore cannot store snowfall or release meltwater. Regional studies have already coupled snow and related cold-region processes to XAJ (Tan et al., 2023; Ju et al., 2024; Dong et al., 2024; Wu et al., 2025), and a large-sample CAMELS study has already coupled CemaNeige to XAJ under differentiable parameter learning across 531 catchments (Chen et al., 2025). What remains under-documented in the HESS-facing literature is a diagnosis-first protocol that (i) maps *where* unmodified XAJ-MZ fails along snow/hydroclimate gradients, (ii) tests a deliberately minimal non-ML snow correction under matched budgets, (iii) requires neutrality on snow-free controls, and (iv) publishes an applicability boundary with fairness checks—rather than proposing another high-performing XAJ–snow family.

Here we implement XAJ-Snow: a single-band, CemaNeige-style degree-day layer with two free parameters—degree-day factor K_f (mm °C^{-1} d^{-1}) and cold-content coefficient CTG [-]—placed upstream of an otherwise unchanged XAJ-MZ core. This draft reports an **engineering go/no-go pilot** (not a population inference): using Caravan forcing for two CAMELS basins under an identical Shuffled Complex Evolution–University of Arizona (SCE-UA) calibration against the Kling–Gupta efficiency (KGE) with a matched medium budget (spotpy `rep`=800, `ngs`=15), out-of-sample Nash–Sutcliffe efficiency (NSE) on snow-affected basin 01013500 (fraction of precipitation falling as snow ≈ 0.37) rises from -0.232 (XAJ-MZ) to 0.732 (XAJ-Snow), while KGE rises from 0.210 to 0.776; on low-snow negative-control basin 14306500, NSE changes only from 0.711 to 0.704 (Δ ≈ -0.006). Calibrated snow parameters on 01013500 remain interior (K_f ≈ 3.50, CTG ≈ 0.116).

Pending (do not treat as completed Results): stratified multi-basin sample; snow-water-equivalent (SWE) auxiliary consistency; optimizer-budget / parameter-freedom factorial that would test whether `rep`=800 is a fair baseline; cross-region applicability regression. Until those experiments finish, the pilot must not be extrapolated to a global claim.

1 Introduction

Large-sample evaluations have shown that "average skill" hides regime-dependent structural failure (Knoben et al., 2020; Santos et al., 2025). Snow-affected catchments are a recurring stress test: models that cannot partition precipitation into rain and snow, or cannot delay melt, mis-time spring peaks even when annual water balance looks plausible.

The Xin' anjiang model is widely used in humid and semi-humid settings. Variants that add snow or related cold-region processes already exist, including coupled snowmelt–XAJ applications (Tan et al., 2023; Ke et al., 2024), distributed degree-day XAJ with ice/snow melt (Ju et al., 2024), conceptual XAJ with snowmelt plus soil freeze–thaw (Dong et al., 2024), graded complexity GXAJ-S / GXAJ-S-SF with SNOW17 (Wu et al., 2025), and a CAMELS-scale (531 basins) differentiable XAJ–CemaNeige system (dMXAJ) with local and regional parameter learning (Chen et al., 2025). Differentiable XAJ formulations further explore optimizer and parameter-sharing design (Ouyang et al., 2024). **This manuscript does not claim the first XAJ snow extension, nor the first XAJ–CemaNeige coupling, nor the first large-sample XAJ–snow evaluation.** Instead, the intended contribution—once large-sample evidence is complete—is a diagnosis-first narrative: map structural inadequacy along snow and hydroclimate gradients; test a deliberately parsimonious *non-ML* correction under matched classical optimization; require neutrality on snow-free controls; and publish an applicability boundary with fairness checks on calibration budget and extra degrees of freedom (Valéry et al., 2014a,b; Premier et al., 2026; Husic et al., 2025; Santos et al., 2025). Relative to Chen et al. (2025), who demonstrate that CemaNeige plus differentiable parameter learning raises median KGE across CAMELS, our complementary question is when a minimal snow layer is *necessary* for structural adequacy under a falsifiable control design—not whether a high-capacity XAJ–snow learner can achieve high skill. Relative to Wu et al. (2025) and Dong et al. (2024), which deepen process complexity within regional cold basins, the distinctive protocol here is paired minimalism plus negative controls and (pending) fairness / applicability diagnostics.

Research questions guiding the full study:

1. RQ1 — Where does XAJ-MZ fail out of sample as a function of snow influence and related attributes?
2. RQ2 — Does a two-parameter, single-band CemaNeige-style layer selectively improve snow-affected basins while remaining neutral on negative controls?
3. RQ3 — Are gains robust to optimizer budget and the extra two parameters, and can an applicability boundary be predicted from catchment attributes?

The present draft reports the engineering pilot that unlocked the go decision for stratified sampling. Large-sample answers to RQ1–RQ3 remain placeholders below.

2 Data and methods

2.1 Caravan data and the two pilot basins

Forcing and discharge come from the Caravan community dataset (Kratzert et al., 2023). Caravan harmonizes multi-source CAMELS-family basins but its meteorological forcing is not identical to the original CAMELS products; this difference can change conceptual-model skill and must be stated explicitly (Clerc-Schwarzenbach et al., 2024). Potential evapotranspiration (PET) in the local diagnosis uses FAO Penman–Monteith attributes supplied with Caravan.

Pilot basins (roles fixed a priori):

Basin	Role	frac_snow	Area (km²)	Aridity (FAO-PM)	Human footprint
camels_01013500	Snow-affected diagnosis target	≈0.37	2298	0.49	30.6
camels_14306500	Low-snow negative control	≈0.0	859	0.49	34.3

Attribute source: results/diagnostics/basin_alignment_01013500_vs_14306500.md (Caravan attributes). Human activity alone cannot explain the skill contrast (similar footprint; control even slightly higher).

Periods (identical for both models and both basins): training 1985-10-01–1995-09-30; testing 2005-10-01–2014-09-30; warmup length 365 days. Daily variables: precipitation P, PET, discharge Q, and for XAJ-Snow also 2 m air temperature T.

2.2 XAJ-MZ baseline

XAJ-MZ denotes the hydromodel implementation of Xin’ anjiang with Muskingum–Zhao routing and fifteen calibrated parameters (no snow store). All precipitation enters the production module as rainfall-equivalent forcing.

2.3 XAJ-Snow: CemaNeige-style single-band layer

XAJ-Snow prepends a lumped (one elevation band) snow-accounting module inspired by CemaNeige (Valéry et al., 2014b) and related airGR documentation, then calls the same XAJ-MZ core on effective liquid input (rain + melt). Fixed thresholds in this implementation: rain/snow threshold T_s = 0 °C and linear mix half-width T_r = 1 °C. Free snow parameters:

- K_f — degree-day melt factor (mm °C^{-1} d^{-1}), search range [0, 10]; larger values melt faster for a given positive temperature.
- CTG — dimensionless cold-content / thermal-state inertia in [0, 1]; larger CTG increases memory of prior cold conditions and delays melt onset.

Mass symbols used in the module: precipitation P, temperature T, snow water equivalent SWE, thermal state G, melt potential MeltPot, snow-cover factor Gratio, and melt M. A schematic daily update (implementation in `hydromodel/models/snow.py`) is:

$$\text{rain, snow} \leftarrow \text{partition}(P, T; T_s, T_r)$$

$$\text{SWE} \leftarrow \text{SWE} + \text{snow}; \quad G \leftarrow \min(0, \text{CTG}\cdot G + (1-\text{CTG})\cdot T)$$

$$\text{MeltPot} \leftarrow \min(\text{SWE}, \max(0, K_f\cdot T)) \quad \text{only if the pack is isothermal } (G \approx 0)\text{; else } \text{MeltPot} = 0$$

$$\text{Gratio} \leftarrow \min(1, \text{SWE} / \text{G}_{\text{threshold}}); \quad M \leftarrow \min(\text{SWE}, (0.9\cdot\text{Gratio} + 0.1)\cdot\text{MeltPot})$$

We describe the layer as *CemaNeige-style / single-band*, not as a strict airGR reproduction (airGR defaults use multiple elevation bands and different temperature bands). K_f ≈ 3.5 mm °C^{-1} d^{-1} on the snow pilot is interior to the search range and within commonly reported degree-day magnitudes (~2–6), but without SWE constraints it remains an effective parameter rather than a physically identified melt coefficient.

2.4 Calibration and metrics

Optimizer: SCE-UA maximizing KGE on the training window. The local medium protocol uses spotpy SCE-UA settings `rep`=800 and `ngs`=15 for *both* models and both basins (identical budget by configuration). **Important:** this matched budget is a controlled engineering comparison; it is *not* yet demonstrated to be a converged or "fair" optimum versus higher `rep`, multiple seeds, or fixed-snow-parameter ablations. Those fairness checks remain pending and must precede any claim that gains are independent of optimization budget or the two extra degrees of freedom (17 vs 15 parameters). Reported skill uses the independent test window. NSE and KGE are defined in the conventional forms:

$$\text{NSE} = 1 - \Sigma(Q_{\text{sim}}-Q_{\text{obs}})^2 / \Sigma(Q_{\text{obs}}-\bar{Q}_{\text{obs}})^2$$

$$\text{KGE} = 1 - \sqrt{[(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2]}$$

where r is correlation, α a variability ratio, and β a bias ratio (hydromodel implementation). NSE = 1 is perfect; NSE = 0 matches climatological mean squared error; negative NSE is worse than the mean. KGE = 1 is perfect; values near 0 indicate severe degradation of correlation, variability, and/or bias components.

2.5 Negative-control design

Basin 14306500 (frac_snow ≈ 0) is the negative control: a useful snow layer should not create large spurious skill gains when snow storage is irrelevant. Human-footprint attributes are similar across the pair, which previously falsified a pure "human disturbance" explanation for 01013500's poor XAJ-MZ skill.

Pending methods blocks: stratified 0/low/mid/high snow sample construction; SWE auxiliary consistency against ERA5-Land (consistency only, not independent ground validation); optimizer×parameter-freedom factorial; attribute-based ΔKGE applicability models with grouped cross-validation.

3 Results

3.1 Engineering pilot (completed)

Table 1 lists paired test-period metrics read directly from `basins_metrics.csv`.

Table 1. Out-of-sample metrics for the go/no-go pilot (SCE-UA + KGE, rep=800).

Basin	Model	NSE	KGE	RMSE	Bias	Corr
01013500	XAJ-MZ	-0.2321	0.2096	2.2446	0.2776	0.2550
01013500	XAJ-Snow	0.7318	0.7764	1.0473	0.1665	0.9031
14306500	XAJ-MZ	0.7106	0.7815	3.0565	-0.3755	0.8461
14306500	XAJ-Snow	0.7043	0.7795	3.0895	-0.2792	0.8410

Source files: results/xaj_snow_go_nogo/.../evaluation_test/basins_metrics.csv; protocol SCE-UA+KGE, rep=800; train 1985-10-01–1995-09-30; test 2005-10-01–2014-09-30; warmup=365.

On 01013500, XAJ-Snow improves NSE by Δ = +0.964 and KGE by Δ = +0.567. On 14306500, ΔNSE = -0.006 and ΔKGE = -0.002, i.e. within a few thousandths—consistent with a non-inflating negative control. Denormalized snow parameters on

01013500: K_f = 3.5006 mm °C^{-1} d^{-1}, CTG = 0.115624 (interior of [0,10]×[0,1]). Unit tests for the snow module: 8/8 passed (`pytest test/test_snow.py`).

3.2 Pilot figures

Figures 1–5 document the pilot. They support an engineering GO decision and method readiness; they are **not** a substitute for stratified population inference.

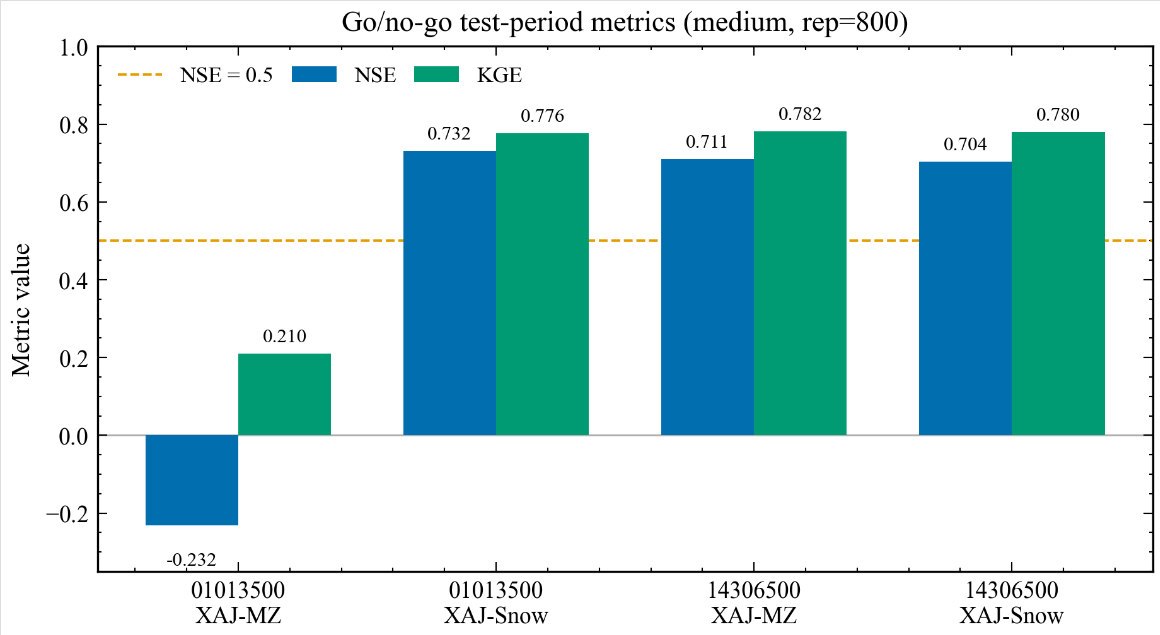

Figure 1. Paired out-of-sample NSE and KGE for the go/no-go pilot basins. Data source: basins_metrics.csv under results/xaj_snow_go_nogo/*/evaluation_test/.

> **Reading Figure 1.** Grouped bars compare NSE (blue family) and KGE (green family) for each basin–model pair. Tall XAJ-Snow bars on 01013500 versus short/negative XAJ-MZ bars show the selective skill recovery; near-equal bars on 14306500 show the control remains flat. The figure cannot prove causality beyond the paired protocol, nor generalize beyond two basins.

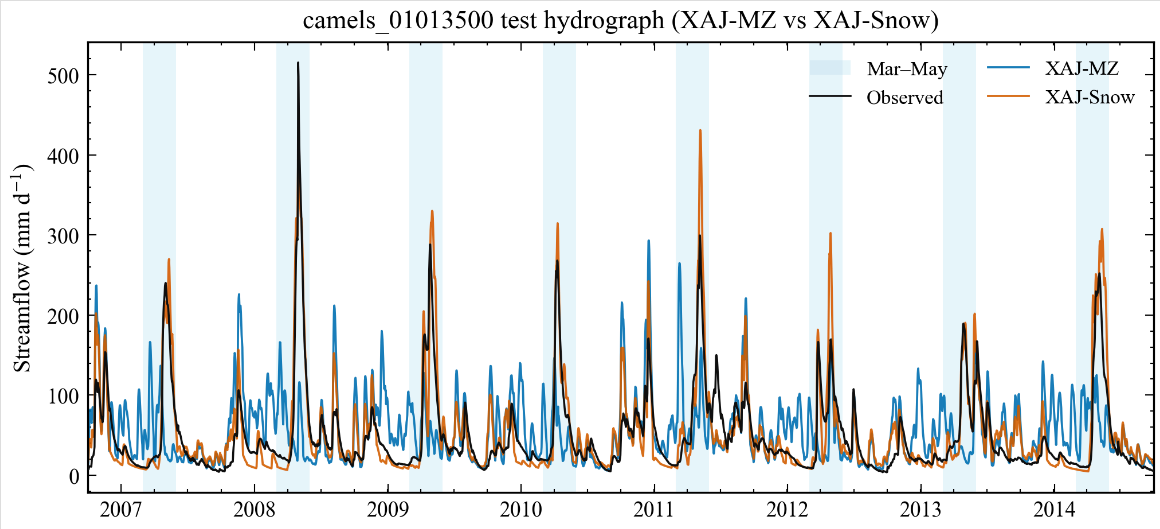

Figure 2. Full test-period hydrograph for snow-affected basin 01013500. Data source: xaj_*_evaluation_results.nc (test window after warmup).

> **Reading Figure 2.** Black: observed discharge; blue: XAJ-MZ; orange: XAJ-Snow over the full test window after warmup. Spring shading marks March–May. XAJ-MZ systematically misplaces snow-season peaks; XAJ-Snow tracks volume and timing more closely. Do not read summer/autumn residuals as snow-process proof; they may reflect soil/routing parameter trade-offs.

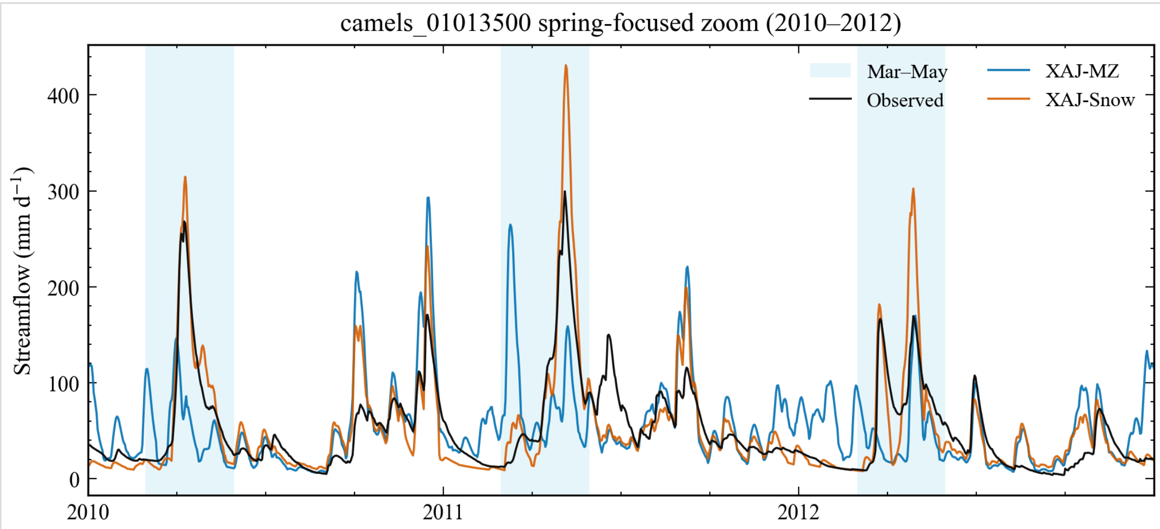

Figure 3. Spring zoom (2010–2012) for basin 01013500. Data source: same NetCDF as Fig. 2; Mar–May spring shading.

> **Reading Figure 3.** A 2010–2012 zoom isolates consecutive snowmelt seasons. Use it to inspect peak timing, multi-peak structure, and whether XAJ-Snow

overshoots individual events. It cannot separate temperature-forcing error from structural melt error.

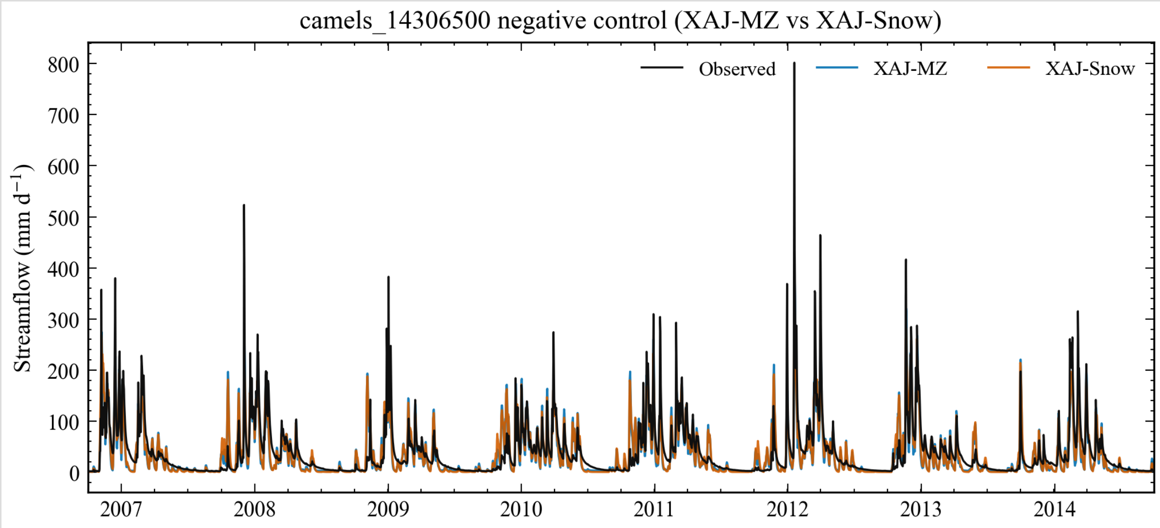

Figure 4. Negative-control hydrograph for low-snow basin 14306500. Data source: xaj_*_evaluation_results.nc for camels_14306500.

Reading Figure 4. On the negative control, XAJ-MZ and XAJ-Snow hydrographs nearly overlap, matching the near-zero Δ metrics. This panel guards against "any extra parameters help everywhere" interpretations.

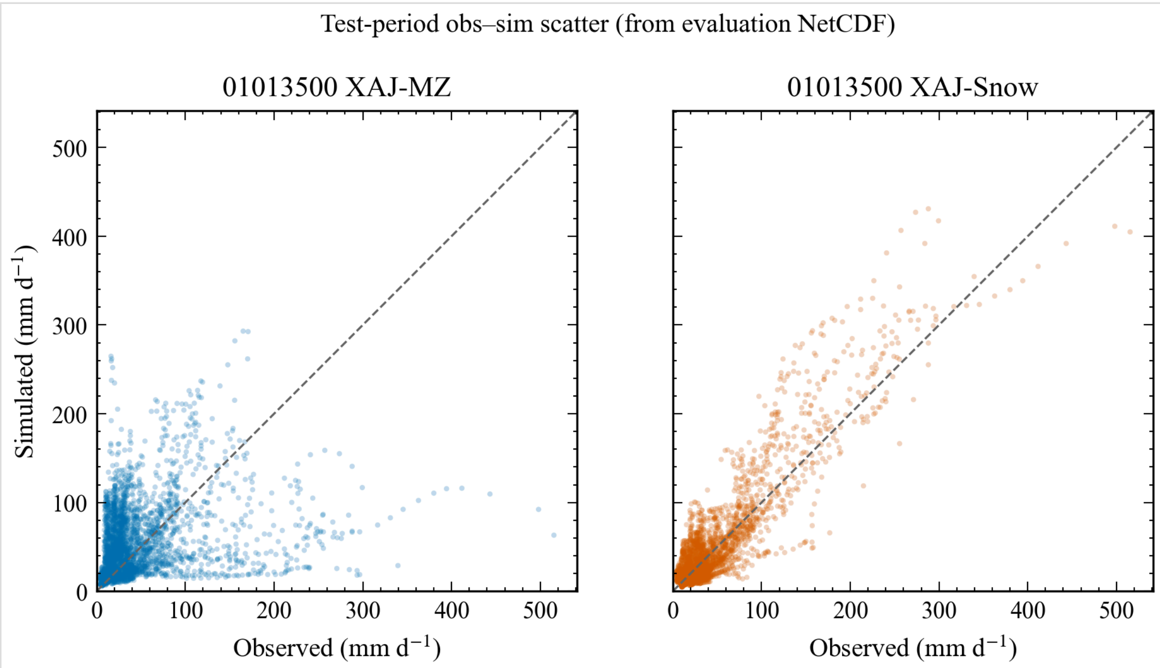

Figure 5. Observed–simulated scatter for basin 01013500. Data source: paired daily Q from evaluation NetCDF.

Reading Figure 5. Daily observed vs simulated scatter for 01013500; the 1:1 line is the reference. XAJ-Snow points hug the diagonal more tightly (higher correlation / lower error), while XAJ-MZ shows larger scatter and bias. Scatter plots compress timing information—pair with Figures 2–3 for hydrograph timing.

Pending Results (fill after experiments):

- Population distribution of XAJ-MZ inadequacy vs snow fraction and covariates.
- Paired ΔNSE/ΔKGE across stratified basins; negative-control cohort statistics.
- SWE auxiliary consistency diagnostics (ERA5-Land), clearly labelled as non-independent.
- Optimizer-budget and fixed-vs-free snow-parameter robustness.
- Applicability boundary model (e.g. GAM / RF+SHAP) with region-grouped CV.
- Optional factorial separating optimizer mechanism from parameter sharing.

4 Discussion

The pilot is consistent with a structural snow-process gap on 01013500 rather than a pure human-activity story: footprint indices are comparable, yet only the snow-affected basin shows large paired gains. The near-zero change on 14306500 reduces—but does not eliminate—concern that seventeen versus fifteen parameters alone buy universal skill; a single negative control cannot replace a stratified zero-snow cohort or a fixed-snow-parameter ablation.

Relative to Tan et al. (2023), Ju et al. (2024), Dong et al. (2024), Wu et al. (2025), and especially Chen et al. (2025), the distinctive claim we aim for is not novelty of "XAJ + snow / CemaNeige", but a multi-regional failure→minimal fix→boundary protocol with explicit negative controls and fairness checks. Chen et al. (2025) already show large-sample skill gains from CemaNeige and from differentiable parameter learning on CAMELS; that result narrows—but does not erase—the niche for a falsifiable diagnosis of *when* a minimal snow store is structurally required under classical SCE-UA and snow-free controls. Wu et al. (2025) remain the closest HESS-facing complexity ladder (GXAJ→GXAJ-S→GXAJ-S-SF); Dong et al. (2024) further add freeze–thaw in a Chinese regional XAJ setting. Our contribution is complementary: parsimony, paired controls, and (pending) applicability / fairness diagnostics—not a competing claim of richer cold-region physics or higher median skill. Relative to Premier et al. (2026), future work should isolate melt-factor effects more tightly (e.g. freeze non-snow parameters) once the stratified sample exists.

Alternative explanations that remain open: Caravan forcing biases (Clerc-Schwarzenbach et al., 2024); PET product choice; single-band temperature representativeness; equifinality among soil parameters absorbing snow errors; and residual anthropogenic regulation not captured by footprint indices. SWE was not used as a calibration target here; without snow-state constraints, K_f and CTG remain effective parameters (Ruelland, 2023, 2024; Tong et al., 2022). Interior K_f/CTG values support numerical stability of the search but do not alone prove physical melt identification.

5 Conclusions

Under a paired SCE-UA+KGE protocol, a two-parameter CemaNeige-style layer converts a strongly negative out-of-sample NSE on snow-affected basin 01013500 into a clearly positive score, while leaving a low-snow control essentially unchanged. This supports an engineering GO for stratified large-sample XAJ-Snow experiments. It does *not* yet establish a global applicability map, independent SWE validation, or optimizer-versus-complexity attribution.

Next steps: freeze Caravan version and screening rules; run stratified batches; add SWE consistency and fairness controls; then rewrite Abstract/Results without pilot-only extrapolation.

Code and data availability

To be completed before submission. Local research code lives in hydromodel 0.3.2 (`hydromodel/models/snow.py`, `xaj_snow.py`). Caravan data follow Kratzert et al. (2023) licensing; local caches under `_portable_data` are not redistributed here. Public repository / DOI / Zenodo package: pending.

Author contributions / Competing interests / Acknowledgements

Author contributions: to be completed.

Competing interests: to be completed (declare none if applicable).

Acknowledgements: to be completed.

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Only locally verified DOIs from docs/local/literature_review_xaj_snow.md are listed.

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Supplementary note on pilot figures

Figure files: `results/figures/fig_*.png` (SciencePlots + Times New Roman, ≥300 dpi). Metric provenance: `results/diagnostics/xaj_snow_go_nogo.md`. This HTML is self-contained (CSS inline; figures as base64). Markdown/PDF siblings are generated by `scripts/generate_publication_outputs.py`.

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